

CHE 311 Homework Week 1

1. Calculate heat flux for the following cases. We only consider the molecular energy transport of heat conduction here.

(1) A block of glycerol with thermal conductivity $k=0.292$ (W/m.k) and a temperature profile of $T = 273 + 3x+4y+5z$ (k).

(2) A single noncubic crystal with $k = \begin{bmatrix} 10 & 5 & 0.38 \\ 5 & 10 & 5 \\ 0.38 & 5 & 10 \end{bmatrix}$ (W/m.k) and a temperature profile of $T = 273 + 3x+4y+5z$ (k).

2. Measurement of heat conductivity

A metal panel of area $A = 100 \text{ cm}^2$ and thickness $Y = 0.5 \text{ cm}$ was found to conduct heat at a rate of 3 W at steady state with temperatures $T_0 = 25^\circ\text{C}$ and $T_1 = 30^\circ\text{C}$ imposed on the two main surfaces. What is the average thermal conductivity of the material at this temperature range?